

Applied Statistics (MATH10096)

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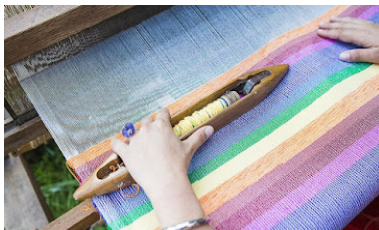


Semester 2, 2024/2025

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- ↪ The `warpbreak` data in R come from an industrial experiment attempting to find optimal conditions for weaving with the minimum number of breaks in the wool being woven.
- ↪ Two types of `wool` (A and B) were tested at each of three weaving `tensions` (**Low**, **Medium**, **High**), with 9 replicates per each combination of wool type and tension level, and the number of `breaks` in a standard length of cloth was recorded.
- ↪ The experiment was designed in such a way that replicates for each combination of wool type and tension level are independent.



Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

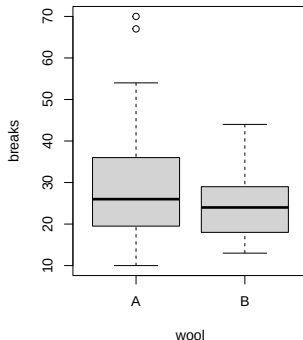
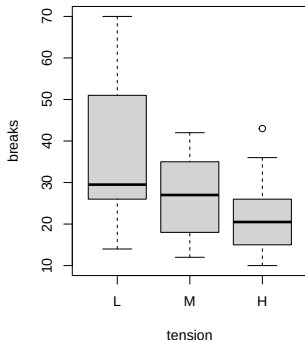
→ Let us start by inspecting the data.

```
## [1] "breaks" "wool" "tension"
## [1] 26 30 54 25 70 52 51 26 67 18 21 29 17 12 18 35 30 36 36 21 24 18 10 43 28
## [26] 15 26 27 14 29 19 29 31 41 20 44 42 26 19 16 39 28 21 39 29 20 21 24 17 13
## [51] 15 15 16 28
## [1] A A A A A A A A A A A A A A A A A A A A A B B B B B B B B B B
## [39] B B B B B B B B B B B B B B B
## Levels: A B
## [1] L L L L L L L L L M M M M M M M M H H H H H H H H L L L L L L L L M M
## [39] M M M M M M M M H H H H H H H H H
## Levels: L M H
```

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

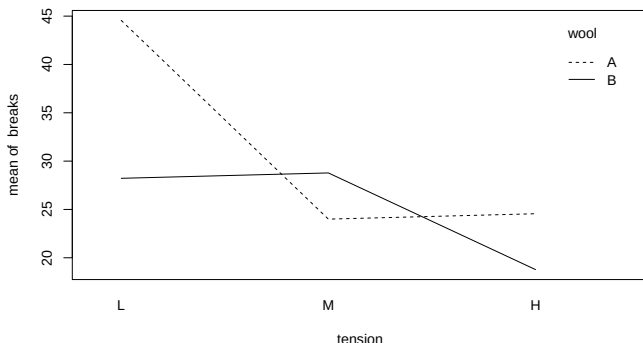
↪ Boxplots of the response variable (number of breaks) against each factor variable (wool type and tension level).



Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- A typical question in this setting is: Does the effect of tension level on the number of breaks depend on the wool type, or vice versa?
- A natural first step to investigate this is the so-called **interaction plot**.



Practical modelling with factors and interactions

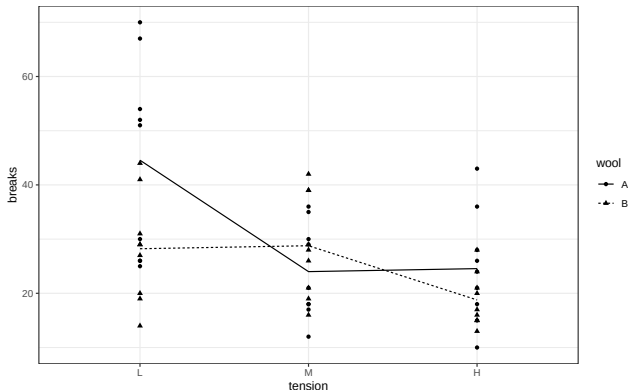
Two factors interacting: the `warpbreak` data

- ↪ What can we conclude from an interaction plot?
 - ↪ If the lines are parallel, or nearly parallel, it suggests little to no interaction between tension level and wool type.
 - ↪ This implies that the effect of wool type remains consistent across all three tension levels, meaning there is no evidence of a differential effect.
 - ↪ If the lines are not parallel, as is the case here, this suggests an interaction between the two factor predictors.
 - ↪ For instance, wool B appears to result in fewer breaks at the lowest tension level; however, this is no longer the case at medium tension.

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

→ A more sophisticated version can be plotted with `ggplot2`.



→ As already evident in one of the boxplots, there is also some evidence of non-constant variation.

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- ↪ Let us therefore consider a model with an interaction between wool type and tension level.
- ↪ Letting y_{ijk} to denote the number of breaks for the k^{th} replicate at the i^{th} wool type and j^{th} tension, the mathematical formulation of the model is

$$y_{ijk} = \mu + \alpha_i + \delta_j + \gamma_{ij} + \epsilon_{ijk}, \quad i \in \{A, B\}, \quad j \in \{L, M, H\}, \quad k \in \{1, \dots, 9\}.$$

- ↪ Here we have that:
 - ↪ The two α_i , $i \in \{A, B\}$, parameters are for the *main effect of wool*.
 - ↪ The three δ_j parameters, $j \in \{L, M, H\}$, are for the *main effect of tension*.
 - ↪ The six γ_{ij} parameters represent the *interaction of wool type and tension*.
 - ↪ μ is the overall expected number of breaks.

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

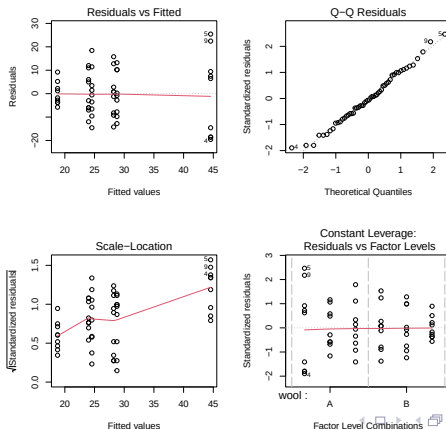
- ↪ Identifiability constraints are needed for this model, of course, but these are generated automatically in `R`.
- ↪ We already know what `R` does by default: set α_A , δ_L , γ_{AL} , γ_{AM} , γ_{AH} , and γ_{BL} to zero.
- ↪ Note that the model actually only has six identifiable parameters (rather than the 12 originally specified).

Practical modelling with factors and interactions

Two factors interacting: the warpbreak data

→ Let us fit the model and check the residuals plots.

```
wm0 <- lm(breaks ~ wool*tension, data = warpbreaks)
par(mfrow = c(2, 2))
plot(wm0)
```

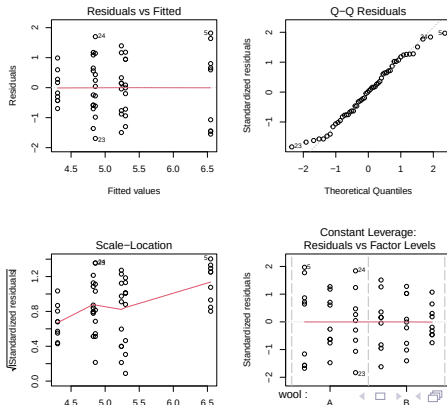


Practical modelling with factors and interactions

Two factors interacting: the warpbreak data

- The variance seems to be increasing with the mean here. Data are counts (number of breaks) and, as such, the square root transformation of the response data is sensible.

```
wm <- lm(sqrt(breaks) ~ wool*tension, data = warpbreaks)
par(mfrow = c(2, 2))
plot(wm)
```



Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- The revised plots show modest improvement, and experimenting with other transformations does not do much better, so the square root transform is probably sensible.

```
summary(wm)

##
## Call:
## lm(formula = sqrt(breaks) ~ wool * tension, data = warpbreaks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.69410 -0.70129  0.01772  0.66046  1.81902
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.5476     0.3266  20.046 < 2e-16 ***
## woolB         -1.3094     0.4619   -2.835 0.006692 **
## tensionM       -1.7216     0.4619   -3.727 0.000511 ***
## tensionH       -1.6912     0.4619   -3.661 0.000625 ***
## woolB:tensionM  1.7821     0.6533    2.728 0.008874 **
## woolB:tensionH  0.7553     0.6533    1.156 0.253350
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9799 on 48 degrees of freedom
## Multiple R-squared:  0.3606, Adjusted R-squared:  0.294
## F-statistic: 5.415 on 5 and 48 DF,  p-value: 0.0004998
```

Practical modelling with factors and interactions

Two factors interacting: the `warpbreaks` data

- ↪ Note that we can also express the identifiable version of the two-way ANOVA model using dummy variables.
- ↪ To this end, let I index a combination of wool type, tension level, and replicate, where $I = 1, \dots, 54$.
- ↪ That is, I represents the index of a row like those shown below.

```
head(warpbreaks)
##      breaks wool tension
## 1       26    A        L
## 2       30    A        L
## 3       54    A        L
## 4       25    A        L
## 5       70    A        L
## 6       52    A        L
```

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

↪ We thus write

$$y_I = \beta_0 + \beta_B \text{wood}_{BI} + \beta_M \text{tension}_{MI} + \beta_H \text{tension}_{HI} + \beta_{BM} \text{wood}_{BI} \text{tension}_{MI} + \beta_{BH} \text{wood}_{BI} \text{tension}_{HI} + \epsilon_I,$$

where

$$\text{wood}_{BI} = \begin{cases} 1, & \text{if the wool type for observation } I \text{ is B,} \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{tension}_{MI} = \begin{cases} 1, & \text{if the tension level for observation } I \text{ is M,} \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{tension}_{HI} = \begin{cases} 1, & \text{if the tension level for observation } I \text{ is H,} \\ 0, & \text{otherwise.} \end{cases}$$

↪ Note that to model the interaction effect, we must consider all possible combinations of wool type and tension levels, except for the reference categories.

↪ Note that we further assume $\epsilon_I \stackrel{iid}{\sim} N(0, \sigma^2)$.

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

↪ Recall the model:

$$y_I = \beta_0 + \beta_B \text{wood}_{BI} + \beta_M \text{tension}_{MI} + \beta_H \text{tension}_{HI} + \beta_{BM} \text{wood}_{BI} \text{tension}_{MI} + \beta_{BH} \text{wood}_{BI} \text{tension}_{HI} + \epsilon_I.$$

↪ The coefficients can be interpreted as follows:

- ↪ β_0 : expected number of breaks if wool type is A (reference) and tension level is L (reference).
- ↪ $\beta_0 + \beta_B$: expected number of breaks if wool type is B and tension level is L (reference).
- ↪ $\beta_0 + \beta_M$: expected number of breaks if wool type is A (reference) and tension level is M.
- ↪ $\beta_0 + \beta_H$: expected number of breaks if wool type is A (reference) and tension level is H.
- ↪ $\beta_0 + \beta_B + \beta_M + \beta_{BM}$: expected number of breaks if wool type is B and tension level is M.
- ↪ $\beta_0 + \beta_B + \beta_H + \beta_{BH}$: expected number of breaks if wool type is B and tension level is H.

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- As always with factor variables, it is not sensible to try and base model selection on the p-values in the summary.
- To perform model selection we can test the null hypothesis that the data are actually described by the model

$$y_I = \beta_0 + \beta_B \text{wood}_{BI} + \beta_M \text{tension}_{MI} + \beta_H \text{tension}_{HI} + \epsilon_I,$$

against the alternative that the model with interactions is required.

- As already alluded to several times, an F-ratio test is the way to do this, as follows.

```
wml <- lm(sqrt(breaks) ~ wool + tension, data = warpbreaks)
anova(wml, wml)

## Analysis of Variance Table
##
## Model 1: sqrt(breaks) ~ wool + tension
## Model 2: sqrt(breaks) ~ wool * tension
##   Res.Df    RSS Df Sum of Sq   F    Pr(>F)
## 1      50 53.291
## 2      48 46.089   2    7.2014 3.75 0.03067 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- ↪ The p-value is quite low here, suggesting that there is evidence to reject the null hypothesis, and accept the alternative that the model with interaction is required.
- ↪ Note that it is meaningless to try and remove main effects while leaving in a corresponding interaction.
- ↪ This mean that we are done, and have selected a model (the original one). So, the way that tension influences the number of breaks is dependent on type of wool.

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- ↪ Having selected the model, which says that the number of breaks depends on an interaction of wool type and tension, we would like to know the combination(s) of wool type and tension that minimize breakages.
- ↪ From the summary of the `wlm` fitted model object this is not easy to determine because of the way that the identifiability constraints have changed the meanings of the remaining parameters.
- ↪ However, we can modify the identifiability constraints in order to get more interpretable parameters. To this end, it is easier to define a new factor variable, say `WT`, whose number of levels equals the product of the two factor levels.

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

→ In order to know the combination(s) of wool type and tension that minimize breakages it is easier to define a new factor variable, say *WT*, whose number of levels equals the product of the two factor levels.

→ Let

$$WT = \begin{cases} AL, & \text{if wool type is A and tension is L,} \\ AM, & \text{if wool type is A and tension is M,} \\ AH, & \text{if wool type is A and tension is H,} \\ BL, & \text{if wool type is B and tension is L,} \\ BM, & \text{if wool type is B and tension is M,} \\ BH, & \text{if wool type is B and tension is H.} \end{cases}$$

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- We know that there are only six identifiable parameters, and so we will let the intercept of the model to be zero. Using dummy variables, the model can then be written as

$$y_I = \beta_{AL} WT_{AL,I} + \beta_{AM} WT_{AM,I} + \beta_{AH} WT_{AH,I} + \beta_{BL} WT_{BL,I} + \beta_{BM} WT_{BM,I} + \beta_{BH} WT_{BH,I} + \epsilon_I.$$

- We define the dummy variable

$$WT_{AL,I} = \begin{cases} 1, & \text{if observation } I \text{ is at level } AL \text{ of variable } WT \text{ (i.e., if for observation } I \text{ wool type } A) \\ 0, & \text{otherwise.} \end{cases}$$

- The other dummy variables $WT_{AM,I}$, $WT_{AH,I}$, $WT_{BL,I}$, $WT_{BM,I}$, and $WT_{BH,I}$ are defined in a similar fashion.
- Under this formulation, β_{AL} is the expected number of breaks for wool type A and tension L, β_{AM} is the expected number of breaks for wool type A and tension M, etc: just what is needed.

Practical modelling with factors and interactions

Two factors interacting: the warpbreak data

↪ We can achieve this in R as follows.

```
wma <- lm(sqrt(breaks) ~ wool:tension - 1, data = warpbreaks)
summary(wma)

##
## Call:
## lm(formula = sqrt(breaks) ~ wool:tension - 1, data = warpbreaks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.69410 -0.70129  0.01772  0.66046  1.81902
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## woolA:tensionL    6.5476     0.3266   20.05  <2e-16 ***
## woolB:tensionL    5.2381     0.3266   16.04  <2e-16 ***
## woolA:tensionM    4.8259     0.3266   14.78  <2e-16 ***
## woolB:tensionM    5.2987     0.3266   16.22  <2e-16 ***
## woolA:tensionH    4.8564     0.3266   14.87  <2e-16 ***
## woolB:tensionH    4.3022     0.3266   13.17  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9799 on 48 degrees of freedom
## Multiple R-squared:  0.9697, Adjusted R-squared:  0.9659
## F-statistic: 255.8 on 6 and 48 DF,  p-value: < 2.2e-16
```

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- ↪ Note the important point that this is not a different model to `wlm`: just a different parameterization of the same model.

```
range(fitted(wlm) - fitted(wma))  
## [1] -7.105427e-15  5.329071e-15
```

- ↪ We can immediately see that wool type B at high tension is estimated to give the lowest break rate.
- ↪ But given the uncertainty can we really be sure that this combination is better than wool A at medium or high tension?
- ↪ Such questions may be important if there are other considerations to take into account when choosing between wool types and tension levels (e.g. cost, wear rate of machinery, etc).

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

- ↪ One obvious thing to do would be to find a confidence interval for the difference in break rate between wool B at high tension and wool A at medium tension, i.e., a confidence interval for $\beta_{AM} - \beta_{BH}$.
- ↪ To do this we need to find the standard deviation of $\hat{\beta}_{AM} - \hat{\beta}_{BH}$.
- ↪ We know that $\widehat{\text{var}}(\hat{\beta}_{AM} - \hat{\beta}_{BH}) = \widehat{\text{var}}(\hat{\beta}_{AM}) + \widehat{\text{var}}(\hat{\beta}_{BH}) - 2\widehat{\text{cov}}(\hat{\beta}_{AM}, \hat{\beta}_{BH})$.
- ↪ From the summary output we have that $\widehat{\text{var}}(\hat{\beta}_{AM}) = \widehat{\text{var}}(\hat{\beta}_{BH}) = 0.3266^2 = 0.106688$.

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

↪ The estimated covariance matrix for the $\hat{\beta}$'s parameters is also easily obtained.

```
vcov(wma)

##                woolA:tensionL woolB:tensionL woolA:tensionM woolB:tensionM
## woolA:tensionL      0.106688      0.000000      0.000000      0.000000
## woolB:tensionL      0.000000      0.106688      0.000000      0.000000
## woolA:tensionM      0.000000      0.000000      0.106688      0.000000
## woolB:tensionM      0.000000      0.000000      0.000000      0.106688
## woolA:tensionH      0.000000      0.000000      0.000000      0.000000
## woolB:tensionH      0.000000      0.000000      0.000000      0.000000
##                woolA:tensionH woolB:tensionH
## woolA:tensionL      0.000000      0.000000
## woolB:tensionL      0.000000      0.000000
## woolA:tensionM      0.000000      0.000000
## woolB:tensionM      0.000000      0.000000
## woolA:tensionH      0.106688      0.000000
## woolB:tensionH      0.000000      0.106688
```

↪ It's diagonal: $\widehat{\text{cov}}(\hat{\beta}_{AM}, \hat{\beta}_{BH}) = 0!$

Practical modelling with factors and interactions

Two factors interacting: the `warpbreak` data

↪ Hence, $\widehat{\text{var}}(\hat{\beta}_{AM} - \hat{\beta}_{BH}) = 2 \times 0.106688 \approx 0.2134$.

↪ The required 95% confidence interval is therefore given by

$$(\hat{\beta}_{AM} - \hat{\beta}_{BH}) \pm t_{0.975, 54-6} \times \sqrt{\widehat{\text{var}}(\hat{\beta}_{AM} - \hat{\beta}_{BH})} = (4.8259 - 4.3022) \pm 2.011 \times \sqrt{0.2134} = 0.52 \pm 0.93$$

↪ Because the confidence interval contains the value zero, we really can't tell apart wool B at high tension and wool A at medium (or high) tension.

Anova tables

Several models argument

- ↪ It is important to know exactly what the entries in these *ANOVA tables* mean, since they are a standard way of displaying results.
- ↪ As seen before, `anova` does different things depending on whether you call it with a single model argument, or several models.
- ↪ Let's consider the multiple model case first, using two of the `warpbreaks` models (with and without the interaction term).

```
anova(wm1, wm)

## Analysis of Variance Table
##
## Model 1: sqrt(breaks) ~ wool + tension
## Model 2: sqrt(breaks) ~ wool * tension
##   Res.Df    RSS Df Sum of Sq   F  Pr(>F)
## 1      50 53.291
## 2      48 46.089   2    7.2014 3.75 0.03067 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Anova tables

Several models argument

```
anova(wm1, wm)

## Analysis of Variance Table
##
## Model 1: sqrt(breaks) ~ wool + tension
## Model 2: sqrt(breaks) ~ wool * tension
##   Res.Df    RSS Df Sum of Sq   F    Pr(>F)
## 1      50 53.291
## 2      48 46.089   2    7.2014 3.75 0.03067 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- ↪ Res.Df gives the **residual degrees of freedom** for each model: i.e., the number of observations (n) less the number of model parameters (p_0 and p_1 , say, for the null (no interaction) and full models (w/ interaction), respectively).
- ↪ RSS gives the **residual sum of squares** for each model (i.e. $RSS_0 = \|\mathbf{y} - \mathbf{X}_0\hat{\beta}_0\|^2$ and $RSS_1 = \|\mathbf{y} - \mathbf{X}_1\hat{\beta}_1\|^2$).
- ↪ Df gives the **difference** in degrees of freedom between each model and the model above it in the table: that is $p_1 - p_0$ in the current case.

Anova tables

Several models argument

```
anova(wm1, wm)

## Analysis of Variance Table
##
## Model 1: sqrt(breaks) ~ wool + tension
## Model 2: sqrt(breaks) ~ wool * tension
##   Res.Df    RSS Df Sum of Sq   F Pr(>F)
## 1      50 53.291
## 2      48 46.089   2    7.2014 3.75 0.03067 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Sum of Sq gives the **difference** in residual sum of squares between each model and the model above it in the table (from 2nd model). In terms of the F-test statistic this is $RSS_0 - RSS_1$.
- F is the **F-ratio test statistic** for comparing each model to the model above it in the table. That is $(RSS_0 - RSS_1) / \{(p_1 - p_0)\hat{\sigma}^2\}$. Note that $\hat{\sigma}^2$ is always calculated from the largest model in the table.
- $\text{Pr}(>F)$ is the **p-value** for testing the null hypothesis that the next model up is correct, against the larger alternative described by this row's model.

Anova tables

Single model argument

↪ The second form of the ANOVA table occurs when only one model is supplied to `anova`.

```
anova(wm)

## Analysis of Variance Table
##
## Response: sqrt(breaks)
##
##      Df Sum Sq Mean Sq F value    Pr(>F)
## wool      1  2.902   2.9019   3.0222 0.088542 .
## tension    2 15.892   7.9458   8.2752 0.000817 ***
## wool:tension 2   7.201   3.6007   3.7500 0.030674 *
## Residuals 48 46.089   0.9602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

↪ The `Residuals` row gives the residual degrees of freedom, residual sum of squares and residual variance estimate, $\hat{\sigma}^2$, for the **full model** (`breaks~wool*tension`).

Anova tables

Single model argument

```
anova(wm)

## Analysis of Variance Table
##
## Response: sqrt(breaks)
##          Df Sum Sq Mean Sq F value    Pr(>F)
## wool          1   2.902    2.9019   3.0222 0.088542 .
## tension        2  15.892    7.9458   8.2752 0.000817 ***
## wool:tension   2   7.201    3.6007   3.7500 0.030674 *
## Residuals     48 46.089    0.9602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- ↪ The `wool:tension` row gives the **difference** in degrees of freedom and residual sum of squares between the full model and a model without the interaction (i.e. `breaks~wool+tension`).
- ↪ The `Mean Sq` column gives the difference in sum of squares divided by the difference in degrees of freedom (an example of an $(RSS_0 - RSS_1)/(p_1 - p_0)$ term in the F test).

Anova tables

Single model argument

```
anova(wm)

## Analysis of Variance Table
##
## Response: sqrt(breaks)
##          Df Sum Sq Mean Sq F value    Pr(>F)
## wool          1   2.902   2.9019   3.0222 0.088542 .
## tension       2  15.892   7.9458   8.2752 0.000817 ***
## wool:tension  2   7.201   3.6007   3.7500 0.030674 *
## Residuals    48  46.089   0.9602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- The F value is then the Mean Sq over $\hat{\sigma}^2$: the F-ratio for testing the null hypothesis that the model without the interaction is adequate, against the alternative that the full model is correct.
- The corresponding p-value is the last entry on the row.

Anova tables

Single model argument

```
anova(wm)

## Analysis of Variance Table
##
## Response: sqrt(breaks)
##           Df Sum Sq Mean Sq F value    Pr(>F)
## wool          1   2.902    2.9019   3.0222 0.088542 .
## tension        2  15.892    7.9458   8.2752 0.000817 ***
## wool:tension   2   7.201    3.6007   3.7500 0.030674 *
## Residuals     48 46.089    0.9602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- ↪ The `tension` row compares the model with wool and tension main effects to a model with just a wool effect. (i.e compares `breaks~wool+tension` to `breaks~wool`).
- ↪ The interpretation of the entries is as for the `wool:tension` row, but note that all the differences are taken between the two models being compared, not between the wool only model and the full model. $\hat{\sigma}^2$ in the F-ratio statistic still comes from the full model/last row of the table, however.

Anova tables

Single model argument

```
anova(wm)

## Analysis of Variance Table
##
## Response: sqrt(breaks)
##          Df Sum Sq Mean Sq F value    Pr(>F)
## wool          1  2.902   2.9019   3.0222 0.088542 .
## tension        2 15.892   7.9458   8.2752 0.000817 ***
## wool:tension    2  7.201   3.6007   3.7500 0.030674 *
## Residuals     48 46.089   0.9602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

↪ The `wool` row compares a model with `wool` only to a model with just an intercept (i.e. `breaks~wool` to `breaks~1`). The interpretation of the entries is as for the previous row.

Anova tables

- ↪ When interpreting ANOVA tables it is important to note a few things things:
 - ↪ If you conclude that the interaction of two factors is significant and therefore can not be dropped from the model, then that means that the two factors concerned are significant, no matter what the p-values are for the rows concerning the factors separately.
 - ↪ p-values for the main effects are only meaningful if their interaction is not significant.
 - ↪ For most models, the ANOVA table will differ depending on the order in which terms are removed from the model: this is because estimators for model terms are generally not independent.
 - ↪ Notable exceptions are models involving factor variables fitted to *balanced* data.
 - ↪ In these cases the terms estimators are generally independent, so that the order in which the the model is reduced is immaterial: the `warpbreaks` data are an example of this.

anova **versus** drop1 functions

- ↪ drop1 is often a more sensible function to use than anova with a single argument.
- ↪ drop1 will **never drop main effects if their interaction is present**, and is always comparing the omission of a single term, to the full model, rather than successively dropping more and more terms.
- ↪ Here it is applied to the `wlm` model.

```
drop1(wlm, test = "F")
## Single term deletions
##
## Model:
## sqrt(breaks) ~ wool * tension
##           Df Sum of Sq    RSS      AIC F value    Pr(>F)
## <none>                 46.089  3.4461
## wool:tension  2       7.2014  53.291  7.2859      3.75 0.03067 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```